CSCE 763: Digital Image Processing

Midterm Exam Study Guide

Organized by: Concept → Formula → Example → Where It Appears

1. Human Vision & Image Formation

Concept: The eye forms images using a lens system. Similar triangles relate object size to image size.

Lens Equation (Similar Triangles):

$$\frac{h_{\text{image}}}{f} = \frac{h_{\text{object}}}{d} \quad \Rightarrow \quad h_{\text{image}} = \frac{f \cdot h_{\text{object}}}{d}$$

where f = focal length, d = distance to object.

Photoreceptors:

- **Cones**: 6–7 million, color vision, concentrated in fovea, daylight (photopic)
- **Rods**: 75–150 million, grayscale, peripheral, low-light (scotopic)

Example: Camera with $f = 35$ mm photographs 50 m tall building at 200 m distance.

$$h_{\text{image}} = \frac{35 \text{ mm} \times 50{,}000 \text{ mm}}{200{,}000 \text{ mm}} = 8.75 \text{ mm}$$

Appears in: Lecture 1–2, HW1 P1 (cone density)

1b. Brightness Perception & Optical Illusions

Weber Ratio: $\frac{\Delta I_c}{I} \approx 0.02$ (JND $\approx 2\%$ of background)

Human eye can discriminate $\sim$50 intensity levels, but this varies with background.

Mach Bands: Perceived bright/dark bands at intensity edges that don't exist in the actual intensity profile. Caused by lateral inhibition in retina.

Simultaneous Contrast: Same gray appears lighter on dark background, darker on light background. The perceived intensity depends on surroundings.

Example: At background intensity $I = 100$, the just-noticeable difference is:

$$\Delta I_c = 0.02 \times 100 = 2$$

You can distinguish $I = 100$ from $I = 102$, but not from $I = 101$.

Appears in: Lecture 1–2

1c. Spatial & Intensity Resolution

Spatial Resolution: Number of pixels per unit length (dpi, pixels/mm).

- Higher spatial resolution → finer detail, larger file
- $M \times N$ image with k bits/pixel requires $M \times N \times k$ bits storage

Intensity Resolution: Number of bits per pixel (k bits → 2^k gray levels).

- $k = 8$ bits → 256 levels (standard grayscale)
- Low bit depth causes **false contouring** (visible bands in smooth gradients)

Isopreference Curves: Tradeoff between spatial and intensity resolution for subjective quality.

Example: A 512×512 image with 8 bits/pixel requires:

$$512 \times 512 \times 8 = 2{,}097{,}152 \text{ bits} = 256 \text{ KB}$$

Reducing to 4 bits/pixel halves storage but may show false contouring.

Appears in: Lecture 2

1d. Interpolation Methods

Concept: Used for image resizing, rotation, and geometric transformations.

Nearest Neighbor: Use value of closest pixel. Fast but causes blocky artifacts.
Bilinear Interpolation: Weighted average of 4 nearest neighbors.

$$v(x, y) = ax + by + cxy + d$$

Smoother than nearest neighbor, slight blurring.
Bicubic Interpolation: Uses 16 neighbors (4×4 grid).

$$v(x, y) = \sum_{i=0}^{3} \sum_{j=0}^{3} a_{ij} x^i y^j$$

Best quality, most computationally expensive.

Quality comparison (for enlargement):

- Nearest neighbor: pixelated edges
- Bilinear: smooth but slightly blurry
- Bicubic: sharpest edges, best for photographs

Appears in: Lecture 2–3

2. Pixel Neighborhoods & Adjacency

Concept: Pixels can be adjacent in different ways depending on which neighbors count.

4-Neighbors $N_4(p)$: pixels at $(x \pm 1, y)$ and $(x, y \pm 1)$ — orthogonal only
8-Neighbors $N_8(p)$: 4-neighbors plus diagonals $(x \pm 1, y \pm 1)$
m-Adjacency: Two pixels p, q with values in V are m-adjacent if:

1. $q \in N_4(p)$, OR
2. $q \in N_D(p)$ AND $N_4(p) \cap N_4(q) \cap V = \emptyset$ (no shared 4-neighbor in V)

Example: For $V = \{1\}$, pixels at $(0, 0) = 1$ and $(1, 1) = 1$ are m-adjacent only if neither $(0, 1)$ nor $(1, 0)$ equals 1. This eliminates path ambiguity.

Appears in: Lecture 2–3, HW1 P2 (adjacency), HW1 P3 (paths), Midterm P1a

3. Distance Metrics

Euclidean: $D_e(p,q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
City-Block (Manhattan): $D_4(p,q) = |x_1 - x_2| + |y_1 - y_2|$
Chessboard: $D_8(p,q) = \max(|x_1 - x_2|, |y_1 - y_2|)$

Example: From $p = (0,0)$ to $q = (4,4)$:

- $D_e = \sqrt{32} = 4\sqrt{2} \approx 5.66$
- $D_4 = 4 + 4 = 8$
- $D_8 = \max(4,4) = 4$

Appears in: Lecture 2–3, HW1 P3 (path lengths)

4. Set Operations on Images

Union: $A \cup B$ — pixels in A OR B
Intersection: $A \cap B$ — pixels in A AND B
Complement: A^c — pixels NOT in A
Difference: $A - B = A \cap B^c$ — pixels in A but not B

Example: Shaded region where exactly A and D overlap but not B or C:

$$(A \cap D \cap B^c \cap C^c)$$

Appears in: Lecture 2–3, HW1 P4 (set expressions), Midterm P2

5. Linear vs Nonlinear Operators

Linear Operator: H is linear if $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$
Examples:

- **Linear**: Averaging, Laplacian, convolution, correlation
- **Nonlinear**: Median, max, min, log transform, gamma correction

Example: Prove median is nonlinear. Let $f_1 = [1,2,3]$, $f_2 = [3,2,1]$.
$\text{median}(f_1) = 2$, $\text{median}(f_2) = 2$, so $\text{median}(f_1) + \text{median}(f_2) = 4$
But $\text{median}(f_1 + f_2) = \text{median}([4,4,4]) = 4 \neq 4$... Actually works here.
Try $f_1 = [1,5,2]$, $f_2 = [2,1,5]$: $\text{median}(f_1) = 2$, $\text{median}(f_2) = 2$, sum $= 4$
$f_1 + f_2 = [3,6,7]$, median $= 6 \neq 4 \quad \Rightarrow$ Nonlinear!

Appears in: Lecture 6–7, HW2 P1 (median nonlinearity)

6. Image Arithmetic Operations

Addition: $g(x,y) = f_1(x,y) + f_2(x,y)$ (noise reduction by averaging)
Subtraction: $g(x,y) = f_1(x,y) - f_2(x,y)$ (change detection)
Multiplication: $g(x,y) = f_1(x,y) \times f_2(x,y)$ (masking, ROI selection)

Example: Defect detection in manufacturing.
Store "golden" reference image f_{ref}. For each new product image f_{new}:

$$\text{defect} = |f_{\text{new}} - f_{\text{ref}}|$$

Nonzero regions indicate missing components. **Requirements**: consistent lighting, alignment, no vibration.

Appears in: Lecture 3–4, HW2 P2 (defect detection)

6b. Image Averaging for Noise Reduction

Model: $g_i(x, y) = f(x, y) + \eta_i(x, y)$ where η is noise with variance σ_η^2
Average of K noisy images:

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^{K} g_i(x, y)$$

Key Result: The variance of the averaged image is:

$$\boxed{\sigma_{\bar{g}}^2 = \frac{1}{K} \sigma_\eta^2}$$

Standard deviation: $\sigma_{\bar{g}} = \frac{\sigma_\eta}{\sqrt{K}}$
As $K \to \infty$, $\bar{g}(x, y) \to f(x, y)$ (noise averages to zero).

Example: To reduce noise standard deviation by half, you need:

$$\frac{\sigma_\eta}{\sqrt{K}} = \frac{\sigma_\eta}{2} \quad \Rightarrow \quad K = 4 \text{ images}$$

To reduce by factor of 10, need $K = 100$ images.

Appears in: Lecture 3–4

7. Intensity Transformations

Log Transform: $s = c \cdot \log(1 + r)$ (compress dynamic range, brighten darks)
Gamma (Power-Law): $s = c \cdot r^\gamma$

- $\gamma < 1$: brightens dark regions (expand darks)
- $\gamma > 1$: darkens image (compress darks)
- $\gamma = 1$: identity

Contrast Stretching: $s = \frac{r - r_{\min}}{r_{\max} - r_{\min}} \cdot (L - 1)$

Example: For $\gamma = 0.4$ on a dark X-ray, dark pixels ($r = 0.1$) map to:

$$s = (0.1)^{0.4} \approx 0.40$$

Brightens the image significantly.

Appears in: Lecture 3–5, Midterm P1b (fireworks at night)

8. Histogram Equalization

Normalized Histogram: $p_r(r_k) = \frac{n_k}{MN}$ where n_k = pixel count at level k
Histogram Equalization:

$$s_k = T(r_k) = (L-1)\sum_{i=0}^{k} p_r(r_i) = (L-1)\cdot \mathrm{CDF}(r_k)$$

Round s_k to nearest integer.

Example: 8-level image with $p_r = [0.1, 0.15, 0.25, 0.2, 0.125, 0.075, 0.0625, 0.0375]$
CDF: $[0.1, 0.25, 0.5, 0.7, 0.825, 0.9, 0.9625, 1.0]$
$s_k = 7 \times$ CDF: $[0.7, 1.75, 3.5, 4.9, 5.775, 6.3, 6.7375, 7]$
Rounded: $[1, 2, 4, 5, 6, 6, 7, 7]$

Appears in: Lecture 5–6, HW2 P3 (idempotency), HW2 P4 (discrete matching)

9. Histogram Specification (Matching)

Goal: Transform image with PDF $p_r(r)$ to have PDF $p_z(z)$
Method (use equalization as a bridge):

1. Equalize input: $s = T_r(r) = (L-1)\int_0^r p_r(w)\,dw$
2. Equalize desired: $s = T_z(z) = (L-1)\int_0^z p_z(w)\,dw$
3. Set equal and solve: $T_r(r) = T_z(z) \Rightarrow z = T_z^{-1}(T_r(r))$

Example: Given $p_r(r) = \frac{4r}{3(L-1)^2}$ and $p_z(z) = \frac{2z}{(L-1)^2}$
$T_r(r) = \frac{2r^2}{3(L-1)}, \quad T_z(z) = \frac{z^2}{L-1}$
Setting equal: $\frac{2r^2}{3(L-1)} = \frac{z^2}{L-1} \Rightarrow z = \sqrt{\frac{2}{3}}\,r$

Appears in: Lecture 5–6, HW2 P5 (continuous), Midterm P4

10. Spatial Filtering — Smoothing

Averaging Filter: $\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$ (blur, reduce noise)
Gaussian Filter: $G(x,y) = \frac{1}{2\pi\sigma^2}e^{-(x^2+y^2)/2\sigma^2}$ (smooth blur)
Median Filter: Output = median of neighborhood (removes salt-and-pepper)

Example: Neighborhood $[15, 20, 255, 18, 22, 19, 21, 17, 0]$
Arithmetic mean: $(15+20+255+18+22+19+21+17+0)/9 = 43$
Median: Sort $\rightarrow [0, 15, 17, 18, 19, 20, 21, 22, 255]$, median $= 19$

Appears in: Lecture 6–7, HW3 P1 (blurring histograms), HW3 P3 (bar merging)

11. Spatial Filtering — Sharpening

Laplacian (4-neighbor): $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$ (8-neighbor): $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

Sharpened Image: $g(x, y) = f(x, y) - \nabla^2 f(x, y)$ (subtract Laplacian)

High-Boost: $g = A \cdot f - \text{lowpass}(f) = (A - 1)f + \text{highpass}(f)$

Sobel (horizontal edges): $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ (vertical): $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

Example: Filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

This is a sharpening filter with inversion (all negative). Center weight $\gg$ neighbors $\Rightarrow$ edge enhancement.

Appears in: Lecture 6–7, HW4 P1 (highboost k=3), Midterm P3

12. Convolution vs Correlation

Correlation: $w(x, y) \star f(x, y) = \sum_s \sum_t w(s, t) \cdot f(x + s, y + t)$

Convolution: $w(x, y) * f(x, y) = \sum_s \sum_t w(s, t) \cdot f(x - s, y - t)$

Key difference: Convolution flips the filter 180°. For symmetric filters, they're the same.

Property: If filter coefficients sum to zero, output sum is also zero.

Zero Padding: Extend image with zeros at boundaries for full output.

Example: Position $(0, 0)$ with zero-padded neighborhood $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 5 \\ 0 & 7 & 18 \end{bmatrix}$

Filter $\frac{1}{9}\begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$:

$\frac{1}{9}[(-1)(0 + 0 + 0 + 0 + 5 + 0 + 7 + 18) + (-17)(3)] = \frac{1}{9}[-30 - 51] = -9$

Appears in: Lecture 6–7, HW3 P2 (properties), HW3 P5 (full output), Midterm P3b

13. Fourier Transform

Continuous FT: $F(u) = \int_{-\infty}^{\infty} f(t)\, e^{-j2\pi ut}\, dt$

Rectangular Pulse Formula (height h, from a to b):

$$\boxed{\int_a^b h\, e^{-j2\pi ut}\, dt = \frac{h}{j2\pi u}\left(e^{-j2\pi ua} - e^{-j2\pi ub}\right)}$$

Sifting Property: $\int_{-\infty}^{\infty} f(t)\, \delta(t - t_0)\, dt = f(t_0)$

Example: $\int_{-\infty}^{\infty} (t^2 + 3t + 5)\, \delta(t - 2)\, dt = (2)^2 + 3(2) + 5 = 15$

Appears in: Lecture 7–8, HW4 P2 (piecewise FT), Midterm Bonus

14. Discrete Fourier Transform

1D DFT: $F(u) = \sum_{x=0}^{M-1} f(x)\, e^{-j2\pi ux/M}$
Inverse: $f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u)\, e^{j2\pi ux/M}$
2D DFT: $F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)\, e^{-j2\pi(ux/M+vy/N)}$
Convolution Theorem: $f * h \leftrightarrow F \cdot H$ (convolution $\rightarrow$ multiplication)

Appears in: Lecture 7–8

15. Frequency Domain Filters

Ideal Lowpass: $H(u, v) = \begin{cases} 1 & D(u, v) \leq D_0 \\ 0 & D(u, v) > D_0 \end{cases}$ (causes ringing)
Butterworth Lowpass: $H(u, v) = \frac{1}{1+[D(u,v)/D_0]^{2n}}$ (smooth rolloff)
Gaussian Lowpass: $H(u, v) = e^{-D^2(u,v)/(2D_0^2)}$ (no ringing)
Highpass: $H_{\mathrm{HP}} = 1 - H_{\mathrm{LP}}$
where $D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2}$

Appears in: Lecture 8+

16. Noise Models

Gaussian: $p(z) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(z-\mu)^2}{2\sigma^2}\right]$
Salt-and-Pepper: $p(z) = \begin{cases} P_a & z = a \text{ (black)} \\ P_b & z = b \text{ (white)} \\ 0 & \text{otherwise} \end{cases}$
Uniform: $p(z) = \frac{1}{b-a}$ for $a \leq z \leq b$

Best filters for each noise type:
- Gaussian noise $\rightarrow$ Arithmetic mean, Gaussian smoothing
- Salt-and-pepper $\rightarrow$ Median filter
- Uniform noise $\rightarrow$ Arithmetic mean

Appears in: Lecture 8+, HW4 P3 (10 filter types)

17. Restoration Filters (Mean Filters)

Arithmetic Mean: $\hat{f} = \frac{1}{mn}\sum g(s,t)$
Geometric Mean: $\hat{f} = [\prod g(s,t)]^{1/mn}$ (less blurring than arithmetic)
Harmonic Mean: $\hat{f} = \frac{mn}{\sum 1/g(s,t)}$ (good for salt noise)
Contraharmonic Mean: $\hat{f} = \frac{\sum g(s,t)^{Q+1}}{\sum g(s,t)^{Q}}$ ($Q > 0$: pepper, $Q < 0$: salt, $Q = 0$: arithmetic)
Median: $\hat{f} = \text{median}\{g(s,t)\}$ (best for salt-and-pepper)
Max: $\hat{f} = \max\{g(s,t)\}$ (removes pepper/dark noise)
Min: $\hat{f} = \min\{g(s,t)\}$ (removes salt/bright noise)
Midpoint: $\hat{f} = \frac{1}{2}(\max + \min)$ (Gaussian + uniform noise)
Alpha-Trimmed Mean: Delete $d/2$ lowest and $d/2$ highest, average rest

Example: Neighborhood $[15, 20, 255, 18, 0, 22, 19, 21, 17]$, $d = 2$
Sorted: $[0, 15, 17, 18, 19, 20, 21, 22, 255]$
Trim 1 lowest (0) and 1 highest (255): $[15, 17, 18, 19, 20, 21, 22]$
Alpha-trimmed mean: $(15 + 17 + 18 + 19 + 20 + 21 + 22)/7 = 18.86$

Appears in: Lecture 8+, HW4 P3 (all 10 filter types)

18. Motion Blur & Wiener Filter

Motion Blur Model: $H(u,v) = \int_0^T e^{-j2\pi[u\,x_0(t)+v\,y_0(t)]}\,dt$
For uniform horizontal motion $x_0(t) = at/T$:

$$H(u,v) = \frac{T}{\pi ua}\sin(\pi ua)\,e^{-j\pi ua}$$

Wiener Filter: $\hat{F}(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2+K} \cdot G(u,v)$
where $K = S_\eta/S_f$ (noise-to-signal power ratio).

Example: Given $H(u,v) = 2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2+v^2)}$ and ratio K:

$$\hat{F} = \frac{2\pi\sigma^2 e^{-2\pi^2\sigma^2(u^2+v^2)}}{4\pi^2\sigma^4 e^{-4\pi^2\sigma^2(u^2+v^2)} + K} \cdot G$$

Appears in: Lecture 8+, HW4 P4 (motion blur), HW4 P5 (Wiener)

19. Order of Linear Operations

Key Property: Linear operations (convolutions) are **commutative**.
If h_1 and h_2 are linear filters:

$$(f * h_1) * h_2 = (f * h_2) * h_1 = f * (h_1 * h_2)$$

Result: Averaging then Laplacian = Laplacian then Averaging (same result)

Example: Apply averaging (noise reduction) then Laplacian (edge detection).
Since both are linear, order doesn't matter mathematically. However, averaging first is often preferred in practice to reduce noise before edge detection.

Appears in: Lecture 6–7, HW3 P4 (averaging + Laplacian order)

Quick Reference: Common 3×3 Filters

Filter	Kernel	Effect
Averaging (box)	$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\1 & 1 & 1\\1 & 1 & 1\end{bmatrix}$	Blur, noise reduction
Weighted avg.	$\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\2 & 4 & 2\\1 & 2 & 1\end{bmatrix}$	Smoother blur (Gaussian approx.)
Laplacian (4-conn)	$\begin{bmatrix}0 & -1 & 0\\-1 & 4 & -1\\0 & -1 & 0\end{bmatrix}$	Edge detection
Laplacian (8-conn)	$\begin{bmatrix}-1 & -1 & -1\\-1 & 8 & -1\\-1 & -1 & -1\end{bmatrix}$	Edge detection (diagonal)
Sobel G_x	$\begin{bmatrix}-1 & 0 & 1\\-2 & 0 & 2\\-1 & 0 & 1\end{bmatrix}$	Vertical edge detection
Sobel G_y	$\begin{bmatrix}-1 & -2 & -1\\0 & 0 & 0\\1 & 2 & 1\end{bmatrix}$	Horizontal edge detection
High-boost ($A = 2$)	$\frac{1}{9}\begin{bmatrix}-1 & -1 & -1\\-1 & 17 & -1\\-1 & -1 & -1\end{bmatrix}$	Sharpen (enhance edges)
Prewitt G_x	$\begin{bmatrix}-1 & 0 & 1\\-1 & 0 & 1\\-1 & 0 & 1\end{bmatrix}$	Vertical edge (simpler)